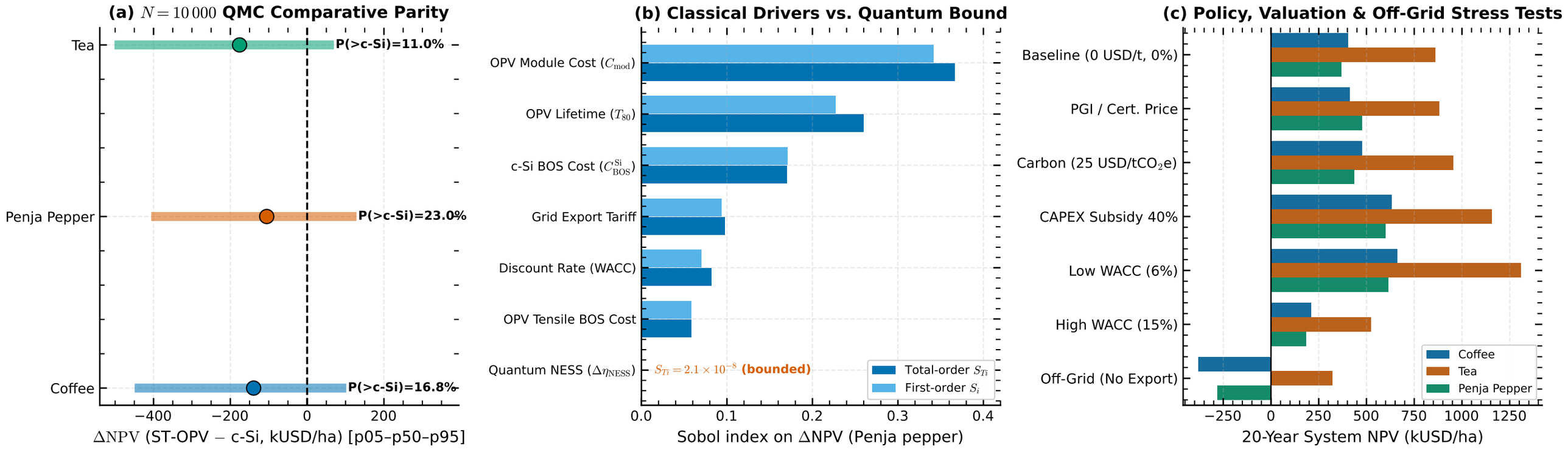

(a) $N = 10\,000$ QMC Comparative Parity
Tea
Penja Pepper
Coffee
P(>c-Si)=11.0%
P(>c-Si)=23.0%
P(>c-Si)=16.8%
−400
−200
0
200
ΔNPV (ST-OPV − c-Si, kUSD/ha) [p05–p50–p95]
(b) Classical Drivers vs. Quantum Bound
OPV Module Cost (C_{mod})
OPV Lifetime (T_{80})
c-Si BOS Cost ($C_{\mathrm{BOS}}^{\mathrm{Si}}$)
Grid Export Tariff
Discount Rate (WACC)
OPV Tensile BOS Cost
Quantum NESS ($\Delta\eta_{\mathrm{NESS}}$)
$S_{Ti} = 2.1 \times 10^{-8}$ (bounded)
Total-order S_{Ti}
First-order S_i
0.0
0.1
0.2
0.3
0.4
Sobol index on ΔNPV (Penja pepper)
(c) Policy, Valuation & Off-Grid Stress Tests
Baseline (0 USD/t, 0%)
PGI / Cert. Price
Carbon (25 USD/tCO$_2$e)
CAPEX Subsidy 40%
Low WACC (6%)
High WACC (15%)
Off-Grid (No Export)
Coffee
Tea
Penja Pepper
−250
0
250
500
750
1000
1250
20-Year System NPV (kUSD/ha)